UNIT – III : EARTH PRESSURE THEORIES & RETAINING WALLS

1. Introduction – Lateral Earth Pressure

- Retaining structures (walls, sheet piles, abutments, basement walls, bunkers) hold back soil → soil pushes **sideways** on them. This horizontal thrust = lateral earth pressure.
- Magnitude depends on: type of soil (c, φ, γ), wall movement (yielding / rigid), wall friction (δ), backfill slope, surcharge, water table, drainage.
- Pressure intensity at depth z : $\sigma_h = K \cdot \sigma_v$, K = coefficient of earth pressure.

2. Three States of Earth Pressure

State	Wall movement	Coefficient
At-rest	None (rigid, unyielding wall) – zero lateral strain	K_0
Active	Wall moves away from soil → soil expands, shear strength fully mobilised	K_a (minimum)
Passive	Wall pushed towards soil → soil compressed	K_p (maximum)

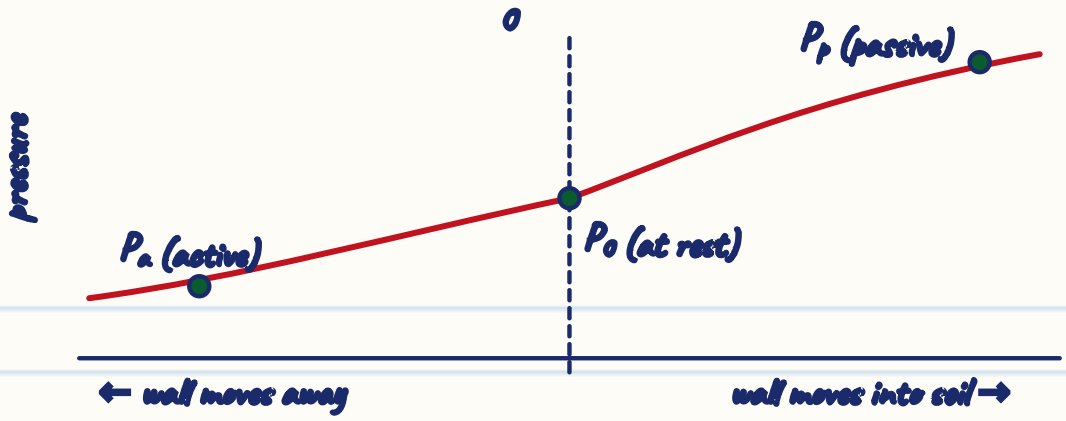


Fig 1 : Lateral pressure vs wall movement

- Movement needed to mobilise: **active** ≈ 0.001 H (dense sand) – 0.004 H (loose sand), **passive** ≈ 0.01 H – 0.05 H (passive needs ≈ 10 times more movement).
- Order : $K_a < K_0 < K_p$

3. Coefficient of Earth Pressure At Rest (K_0)

$$K_0 = \sigma_h/\sigma_v = \nu/(1-\nu) \quad \text{(elastic)}$$
$$K_0 = 1 - \sin\varphi' \text{ (Jaky, N.C. soil)}; \; K_{0(OC)} = K_{0(NC)}\sqrt{OCR}$$

- Typical K_0: loose sand 0.5, dense sand 0.4, N.C. clay 0.5–0.65, over-consolidated clay up to 1.0 or more.
- Pressure at rest: $p_0 = K_0 \gamma z$; thrust $P_0 = \tfrac{1}{2} K_0 \gamma H^2$ acting at $H/3$ from base.

4. Rankine's Theory of Earth Pressure (1857)

Assumptions

- Soil is **homogeneous, isotropic, semi-infinite** and cohesionless (extended later to c-φ soil).
- Wall is **vertical, smooth** (no wall friction, $\delta = 0$) and rigid; the back of the wall does not disturb the stress field.
- Ground surface is plane; plastic equilibrium exists throughout the backfill; for level ground the thrust is **horizontal**.

Active case – cohesionless soil, horizontal backfill

- Vertical stress $\sigma_v = \gamma z$ = **major** principal stress σ_1; horizontal stress $\sigma_h = \sigma_3$ (minor).
- At failure, Mohr circle touches the envelope $\tau = \sigma \tan\varphi$:

$$\sin\varphi = (\sigma_1 - \sigma_3)/(\sigma_1 + \sigma_3) \Longrightarrow \sigma_3 = \sigma_1 (1 - \sin\varphi)/(1 + \sin\varphi)$$

$$\sigma_a = K_a \gamma z, \quad K_a = (1 - \sin\varphi)/(1 + \sin\varphi) = \tan^2(45^\circ - \varphi/2)$$

$$P_a = \tfrac{1}{2} K_a \gamma H^2 \text{ acting at } H/3 \text{ above base}$$

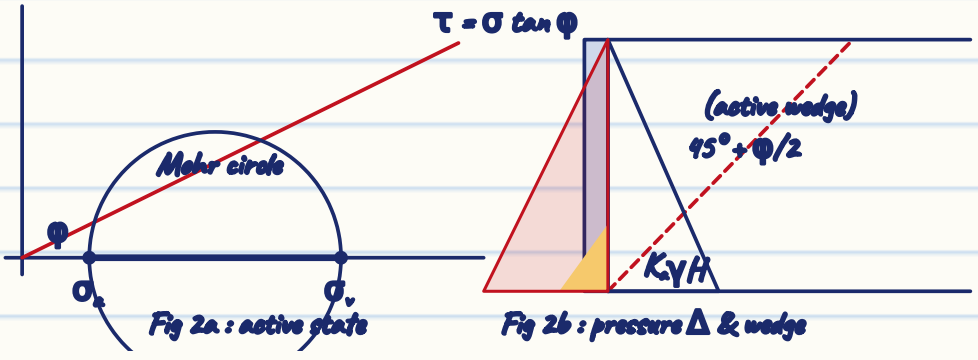


Fig 2a : active state Fig 2b : pressure Δ & wedge

- Failure (slip) planes make $(45° + \varphi/2)$ with the horizontal in the active state; two sets of planes inclined at $\pm (45° + \varphi/2)$.

Passive case – cohesionless soil, horizontal backfill

- Now σ_h becomes the **major** principal stress, $\sigma_v = \gamma z$ the minor.

$$\sigma_p = K_p \gamma z, \quad K_p = (1 + \sin\varphi)/(1 - \sin\varphi) = \tan^2(45° + \varphi/2) = 1/K_a$$

$$P_p = \tfrac{1}{2} K_p \gamma H^2 \text{ at } H/3 \text{; slip planes at } (45° - \varphi/2) \text{ to horizontal}$$

For $\varphi = 30°$: $K_a = 1/3$, $K_p = 3$, $K_0 = 0.5$

5. Rankine's Theory – Cohesive (c–φ) Soil

- Mohr–Coulomb : $\tau = c + \sigma \tan\varphi$. From the Mohr circle:

$$\sigma_a = \gamma z K_a - 2c\sqrt{K_a} \quad / \quad \sigma_p = \gamma z K_p + 2c\sqrt{K_p}$$

- At the surface ($z = 0$) active pressure $= -2c\sqrt{K_a}$ (tension) → soil tends to separate from the wall → **tension cracks.**
- Depth of tension crack ($\sigma_a = 0$):

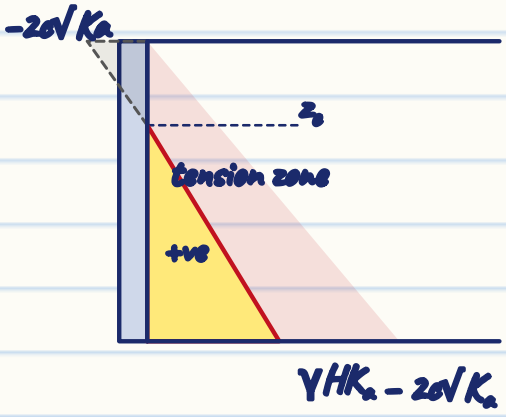
$$z_c = 2c / (\gamma \sqrt{K_a}) = (2c/\gamma) \tan(45° + \varphi/2)$$

- Depth up to which unsupported vertical cut stands (σ_a net = 0 over H): $H_c = 2 z_c = 4c/(\gamma\sqrt{K_a})$.
- For $\varphi = 0$ clay : $K_a = 1 \Longrightarrow \sigma_a = \gamma z - 2c$; $z_c = 2c/\gamma$.

Total active thrust (c–φ soil, height H)

$$\text{Without crack : } P_a = \tfrac{1}{2}\gamma H^2 K_a - 2cH\sqrt{K_a} + 2c^2/\gamma$$

$$\text{With crack : } P_a = \tfrac{1}{2}\gamma K_a (H - z_c)^2 \text{, acts at } (H - z_c)/3 \text{ above base}$$



- Net pressure negative upto z_c
- Pressure diagram: –ve triangle + +ve triangle
- Tension is ignored (soil cannot pull wall) → only +ve part used

Fig 3 : Active pressure in c–φ soil

6. Effect of Uniform Surcharge (q per unit area)

- Surcharge q adds to vertical stress at every depth : $\sigma_v = q + \gamma z$.
- Active pressure : $\sigma_a = K_a(q + \gamma z)$ → trapezoidal diagram. Equivalent height of extra fill $h_e = q/\gamma$.

$$P_a = K_a q H + \tfrac{1}{2} K_a \gamma H^2$$

- Line of action (from base) : $\bar{y} = [K_a q H \cdot (H/2) + \tfrac{1}{2} K_a \gamma H^2 \cdot (H/3)] / P_a$.
- For c–φ soil add $- 2c\sqrt{K_a}$ (constant) to every ordinate.

7. Effect of Water Table (backfill partly / fully submerged)

- Below water table use submerged unit weight $\gamma' = \gamma_{sat} - \gamma_w$ for soil pressure **plus** full hydrostatic pressure $\gamma_w z_w$ (water pressure is not reduced by K).

$$\sigma_{total} = K_a[\gamma H_1 + \gamma'(z - H_1)] + \gamma_w (z - H_1)$$

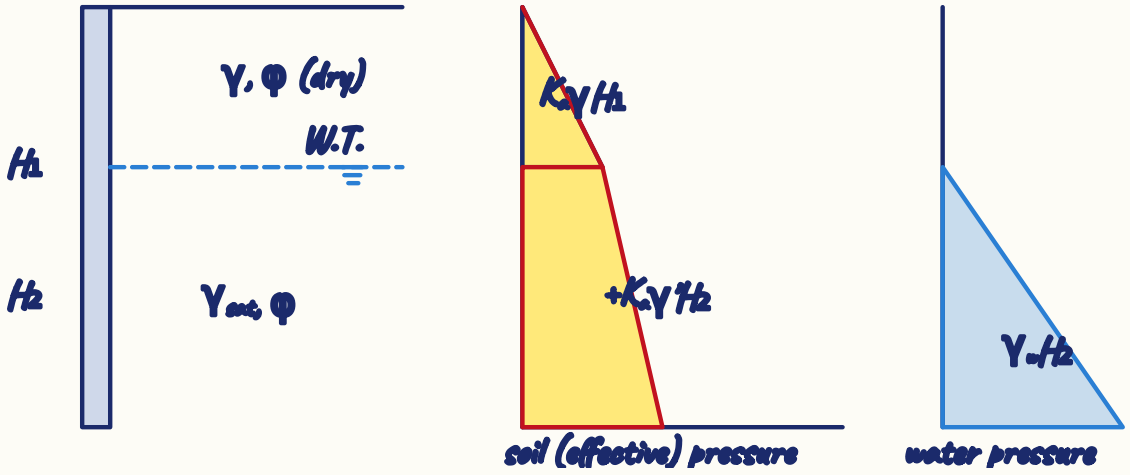


Fig 4 : Pressure diagram with water table at depth H_1

- Thrust = areas of triangle + rectangle + triangle (soil) + $\tfrac{1}{2}\gamma_w H_2^2$ (water). Water increases thrust nearly 2 times → *provide drainage!*
- Passive side with submerged soil: use $K_p \gamma' z + \gamma_w z$ similarly.

8. Inclined Backfill (angle β) – Rankine

$$K_a = \cos\beta \cdot [\cos\beta - \sqrt{(\cos^2\beta - \cos^2\varphi)}] / [\cos\beta + \sqrt{(\cos^2\beta - \cos^2\varphi)}]$$

- $P_a = \tfrac{1}{2} K_a \gamma H^2$ acts **parallel to the sloping surface** (inclined β to horizontal) at H/3. For β = 0 it reduces to $(1 - \sin\varphi)/(1 + \sin\varphi)$; need $\beta \le \varphi$.

9. Earth Pressure in Layered Soils (Rankine)

- Each layer has its own γ, c, φ → its own K_a. Pressure is **continuous in vertical stress σ_v** but jumps at the interface because K changes.
- **Procedure :**
  1. Compute σ_v at each layer boundary (bottom of layer 1 = $\gamma_1 H_1 + q$).
  2. Upper layer: $\sigma_a = K_{a1}\,\sigma_v$ (use above the interface) ; lower layer: $\sigma_a = K_{a2}\,\sigma_v$ (just below the interface).
  3. Below the water table use γ' and add $\gamma_w z_w$. For c-soil subtract $2c\sqrt{K_a}$ in that layer only.
  4. Split diagram into rectangles + triangles; P = areas ; $\bar{y} = \Sigma P{\cdot}y \,/\, \Sigma P$ (moment about base).

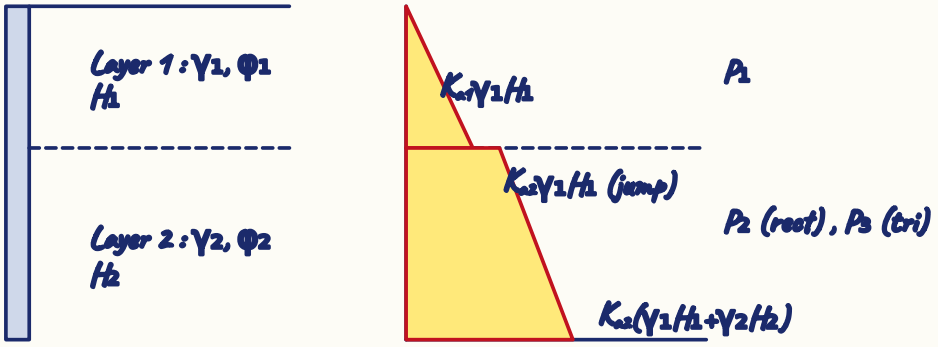


Fig 5 : Two-layer cohesionless backfill

If lower layer is clay ($\varphi = 0$): $K_{a2} = 1 \rightarrow \sigma_a = \sigma_v - 2c$; upper layer acts as surcharge $q = \gamma_1 H_1$.

Example 1 – Layered backfill

Q. A 6 m high smooth vertical wall retains horizontal backfill: 0–3 m γ = 18 kN/m³, φ = 30°; 3–6 m γ = 20 kN/m³, φ = 20°. Find total active thrust and its point of application.

$K_{a1} = (1 - \sin30°)/(1 + \sin30°) = 0.333$; $K_{a2} = (1 - \sin20°)/(1 + \sin20°) = 0.490$

Level	σ_v (kPa)	$\sigma_a = K\,\sigma_v$ (kPa)
0 m	0	0
3 m (above)	18×3 = 54	0.333×54 = 18.0
3 m (below)	54	0.490×54 = 26.5
6 m	54 + 20×3 = 114	0.490×114 = 55.9

Solution (contd.) – Thrusts per metre run & moments about base:

Part	Force (kN/m)	Arm (m)	Moment (kNm)
$P_1 = \frac{1}{2}\times 18\times 3$	27.0	3 + 1 = 4.0	108.0
$P_2 = 26.46\times 3$	79.4	1.5	119.1
$P_3 = \frac{1}{2}\times(55.86-26.46)\times 3$	44.1	1.0	44.1
Total	150.5		271.2

$P_a = 150.5$ kN/m ; $\bar{y} = 271.2/150.5 = 1.80$ m above base

Example 2 – c–φ soil with tension crack

Q. A vertical smooth wall 5 m high retains a clay backfill: $\gamma = 18$ kN/m^3, $c = 10$ kN/m^2, $\varphi = 20°$. Find (a) depth of tension crack (b) active thrust with & without crack, (c) its point of application.

$K_a = \tan^2 35° = 0.490$, $\sqrt{K_a} = 0.700$

- σ_a at top $= -2c\sqrt{K_a} = -2\times 10\times 0.7 =$ **– 14 kPa**
- σ_a at base $= 18\times 5\times 0.49 - 14 = 44.1 - 14 =$ **30.1 kPa**
- (a) $z_c = 2c/(\gamma\sqrt{K_a}) = 20/(18\times 0.7) =$ **1.59 m**
- (b) <u>Without crack</u> (tension allowed) : $P_a = \frac{1}{2}\gamma H^2 K_a - 2cH\sqrt{K_a} = 110.25 - 70 =$ **40.3 kN/m**

  <u>With crack</u> (tension ignored) : $P_a = \frac{1}{2}\times 30.1\times(5-1.59) =$ **51.4 kN/m** ($= \frac{1}{2}\gamma H^2 K_a - 2cH\sqrt{K_a} + 2c^2/\gamma$).
- (c) $\bar{y} = (H - z_c)/3 = 3.41/3 =$ **1.14 m** above base.

Design with the crack (51.4 kN/m) – the safe value. If water fills the crack add $\frac{1}{2}\gamma_w z_c^2$.

10. Rankine vs Coulomb – Why Coulomb ?

- Rankine can't handle **wall friction**, rough/inclined wall backs, irregular backfill → **Coulomb's wedge theory** (1776) can.
- Rankine gives a state of stress (stress-field method, lower bound); Coulomb gives a failure-wedge method (limit equilibrium).

II. Coulomb's Wedge Theory (1776)

Assumptions

- Backfill is dry, **cohesionless**, homogeneous, isotropic; failure surface is a **plane** (BC) – a wedge ABC slides down.
- Wall **rough** : wall friction δ ($\approx$ ⅔φ) acts on back of wall; wall back can be inclined (angle α to horizontal); backfill can slope (β).
- Soil wedge behaves as a **rigid body**; failure is a plane-strain problem; for **active** case wedge moves down-and-out, for passive up-and-in.
- Pa is found by trial wedges and taking the **maximum** value (passive : **minimum**).

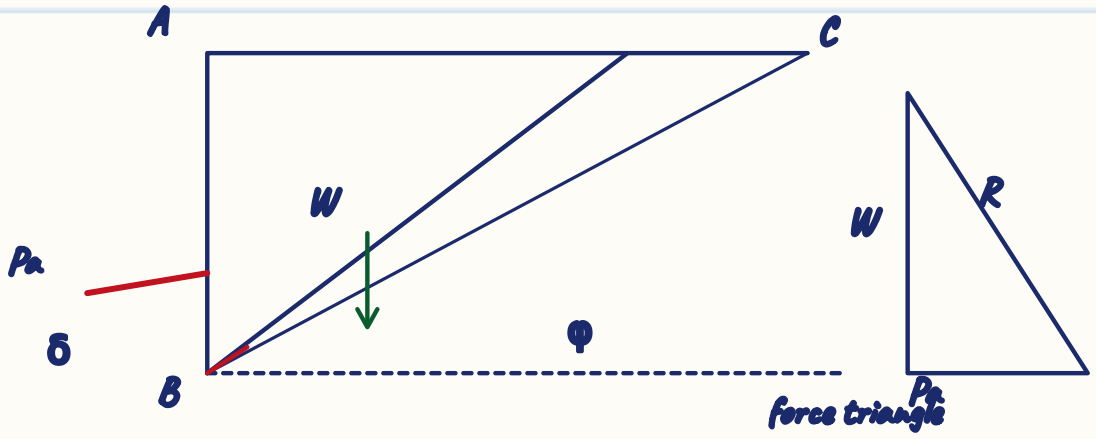


Fig 2 : Coulomb active wedge ABC and its force triangle (W, P_a, R)

- Forces on wedge: **W** (weight, known), P_a (reaction of wall, inclined δ to normal of wall), R (reaction on slip plane, inclined φ to normal). Triangle of forces closes → $P = W \sin(\theta-\varphi)/\sin(\ldots)$

$$P_a = \tfrac{1}{2} K_a \gamma H^2 \text{ (acts at } H/3 \text{ from base, at } \delta \text{ to wall normal)}$$

$$K_a = \sin^2(\alpha+\varphi) \,/\, \{ \sin^2\alpha \cdot \sin(\alpha-\delta) \cdot [1 + \sqrt{\,(\sin(\varphi+\delta)\sin(\varphi-\beta) / (\sin(\alpha-\delta)\sin(\alpha+\beta))\,}\,]^2 \}$$

- α = inclination of wall back with horizontal ; β = backfill slope ; δ = wall friction.
- **Special case** $\alpha = 90°, \delta = 0, \beta = 0 \rightarrow K_a = (1-\sin\varphi)/(1+\sin\varphi)$ = Rankine's K_a ✔.

Coulomb's passive pressure

$$P_p = \tfrac{1}{2} K_p \gamma H^2 \; ; \; K_p = \sin^2(\alpha-\varphi) / \{ \sin^2\alpha \cdot \sin(\alpha+\delta) \cdot [1 - \sqrt{(\sin(\varphi+\delta)\sin(\varphi+\beta) / (\sin(\alpha+\delta)\sin(\alpha+\beta))}\,]^2 \}$$

- For passive case P_p is the **minimum** over trial wedges ; with $\delta > \varphi/3$ the plane-surface assumption over-estimates P_p (unsafe) → use log-spiral surface or take $\delta = 0$ for passive.

Point of application : Coulomb gives only the magnitude; assume it acts at H/3 (for uniform γ, no surcharge). With surcharge → find C.G. of the pressure diagram.

Example 3 – Coulomb K_a

Vertical wall $H = 6$ m, backfill horizontal, $\varphi = 30°$, $\delta = 20°$, $\gamma = 18$ kN/m³. Find P_a.

- $\alpha = 90°$, $\beta = 0$: $\sin^2(120°) = 0.75$; $\sin(\alpha-\delta) = \sin 70° = 0.940$; $\sin(\varphi+\delta) = \sin 50° = 0.766$; $\sin\varphi = 0.5$
- $\sqrt{(0.766 \times 0.5 / 0.940)} = 0.638 \rightarrow [1.638]^2 = 2.684$
- $K_a = 0.75 / (0.940 \times 2.684) = \mathbf{0.297}$ (Rankine gives 0.333)

$$P_a = \tfrac{1}{2} \times 0.297 \times 18 \times 6^2 = 96.3 \text{ kN/m}$$

- $P_h = P_a \cos 20° = 90.5$ kN/m ; $P_v = P_a \sin 20° = 32.9$ kN/m (acts downward at $H/3 = 2$ m above base)

12. Rankine vs Coulomb – Quick Comparison

Rankine	Coulomb
Stress-state (plastic equilibrium) of whole backfill	Limit equilibrium of a rigid wedge
Wall smooth, vertical ($\delta = 0$)	Wall rough, any inclination, $\delta \neq 0$
Gives pressure **distribution** (p vs z)	Gives only total thrust P_a
Applicable to c–φ soil	Cohesionless soil ($c = 0$), dry
Passive – fairly accurate	Passive over-estimated if δ large
Inclined backfill : thrust ∥ ground surface	Thrust at δ to wall normal

13. Culmann's Graphical Method

➤ Graphical version of Coulomb's trial-wedge method – works for **any ground profile, surcharge, layered backfill, c = 0.** Gives P_a and the critical slip plane.

Steps (active case)

1. Draw wall back AB to scale (B = heel, A = top) and the ground line AD.
2. Draw **φ-line** (base line) BD' from B inclined at φ to the horizontal, up to meet the ground line.
3. Draw **pressure line** (ψ-line) from B making angle $\psi = \alpha - \delta$ with the base line (for smooth vertical wall $\psi = 90°$).
4. Choose trial slip planes BC_1, BC_2, BC_3 … ending on the ground; compute weight of each wedge W_1, W_2… (area × γ ; add surcharge load if any).
5. Along the base line from B, mark off W_1, W_2… to a convenient force scale (points 1, 2, 3 …).
6. Through points 1, 2, 3 draw lines **parallel to the pressure line** to meet respective trial planes BC_1, BC_2… at points c_1, c_2…
7. Join c_1, c_2… by a smooth curve = **Culmann's curve.**
8. Draw a **tangent to the curve parallel to the base line;** the point of tangency T gives the critical plane (join B–T and extend).
9. Draw TP ∥ pressure line to cut base line at P : P_a = **length TP × force scale.**

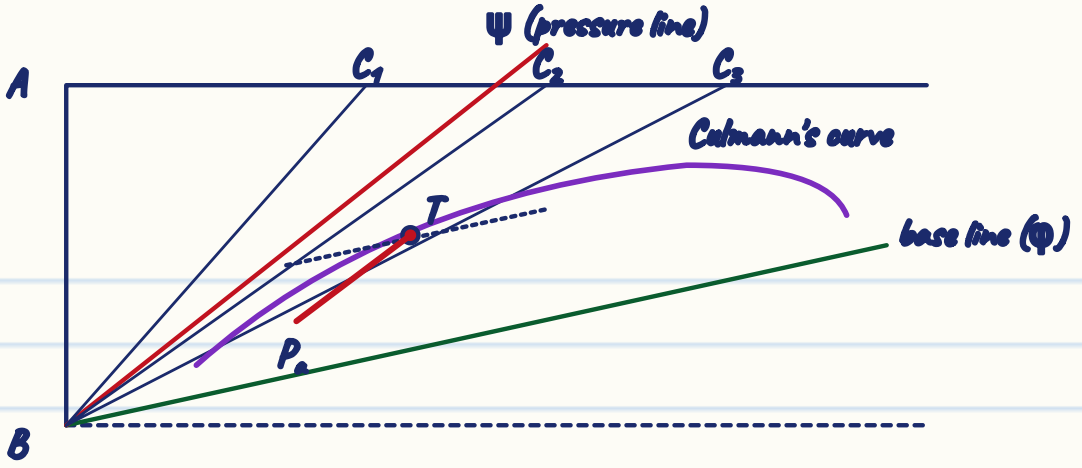


Fig 3 : Culmann's construction (schematic)

Passive case : same procedure but φ-line is drawn below, pressure line on the other side, and the **minimum** ordinate (tangent on the lower side of the curve) gives P_p.

14. Rebhann's (Poncelet's) Graphical Method

- A purely geometrical **construction for Coulomb's wedge** – the critical wedge and P_a are obtained directly, without trial wedges (quicker than Culmann). Only for a **plane ground surface** and uniform soil.

Steps (wall AB, A = top, B = base)

1. Draw wall back AB, ground line AD (slope β), **φ-line** BD from B (angle φ with horizontal) meeting ground line at D.
2. Draw the **pressure (ψ) line** through B as in Culmann ($\psi = \alpha - \delta$ with the φ-line).
3. Through A draw a line $\parallel$ ψ-line to meet BD at C.
4. Draw a **semicircle on BD** as diameter. Through C draw CE $\perp$ BD to meet the semicircle at E.
5. With B as centre and BE as radius cut BD at F.
6. Through F draw FG $\parallel$ ψ-line to meet the ground line AD at G. Join B–G : **BG = critical failure plane.**
7. With F as centre and FG as radius, cut BD at H. Join G–H to form **triangle GHF.**
8. Drop the perpendicular GI on BD (height x). Then

$$P_a = \gamma \times (\text{area of } \triangle GHF) = \tfrac{1}{2}\gamma \cdot HF \cdot x$$

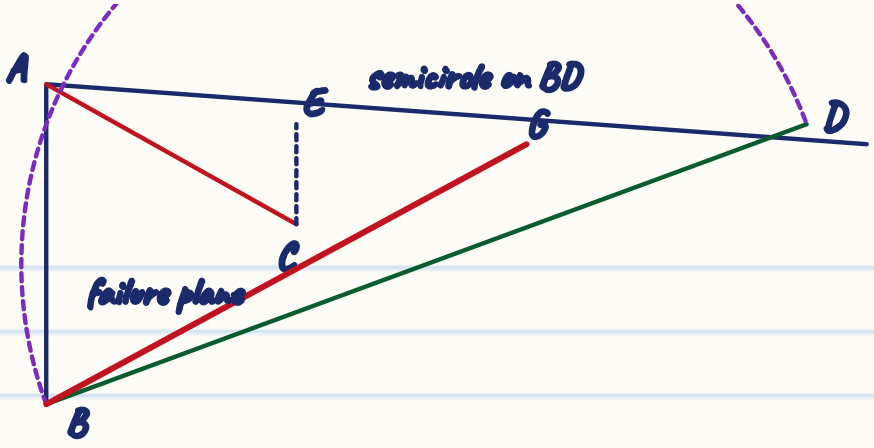


Fig 4 : Rebhann's construction (schematic – lettering as in the steps above)

Exam tip : write the 8 steps in order and draw a neat figure – marks are given for the construction sequence.

15. Active Pressure with Surcharge (Coulomb)

- **Uniform surcharge q** : replace by an equivalent height of soil $h_s = q/\gamma$; $P_a = \frac{1}{2} K_a \gamma H^2 + K_a q H$ (acts at C.G. of trapezoid) – same as Rankine.
- **Line load Q** at distance x : Culmann's method – add Q to W of every wedge that includes the load.

RETAINING WALLS

16. Types of Retaining Walls

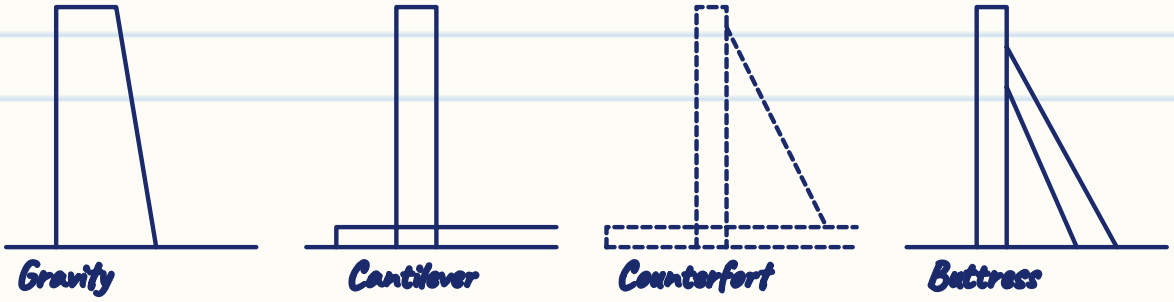


Fig 5 : Common types of retaining walls

Type	Action / Suitable height
Gravity wall (masonry / plain concrete)	Stability from own weight; no steel; H up to ≈ 3–4 m; base width 0.5–0.7 H, top ≥ 0.3 m
Semi-gravity	Gravity wall with light reinforcement near back face; slightly thinner
Cantilever (L / inverted T)	R.C.C. stem + base slab (heel + toe); weight of soil on heel helps stability ; H ≈ 3–7 m; base ≈ 0.4–0.7 H
Counterfort	Vertical ribs inside backfill (in tension) tie stem to heel slab; slab spans horizontally ; H > 6–7 m
Buttress	Ribs on the outer (toe) side, act in compression; more economical but uses space ; H > 6 m
Crib wall / Gabion wall	Interlocking boxes / wire cages filled with stone ; flexible, free-draining; tolerate settlement
Reinforced earth wall	Facing panels + steel / geosynthetic strips in compacted backfill (Henri Vidal) ; very high walls, flexible
Bridge abutment	Retains approach embankment + carries bridge loads
Sheet pile (bulkhead) wall	Cantilever / anchored ; for waterfronts, excavations

17. Drainage of Backfill

- Water adds hydrostatic thrust ($\frac{1}{2}\gamma_w h^2$) → can nearly double the load. Provide: **weep holes** (100 mm ø @ 2–3 m c/c), **filter / graded gravel** behind wall, longitudinal **perforated drain pipe**, surface slope / paving to keep rain out.
- Use **granular (free-draining) backfill** – clay backfill is avoided (swelling, high pressure).

18. Stability of Retaining Walls – Forces

- Forces per metre run : (i) **weight of wall W** (split into rectangles/triangles), (ii) **weight of soil over heel**, (iii) **active thrust P_a** (Rankine for cantilever – on a virtual vertical plane through heel, Coulomb for gravity), (iv) **passive resistance P_p** in front of toe (often neglected – may be scoured), (v) base reaction (N and friction F), (vi) surcharge, water pressure if any.

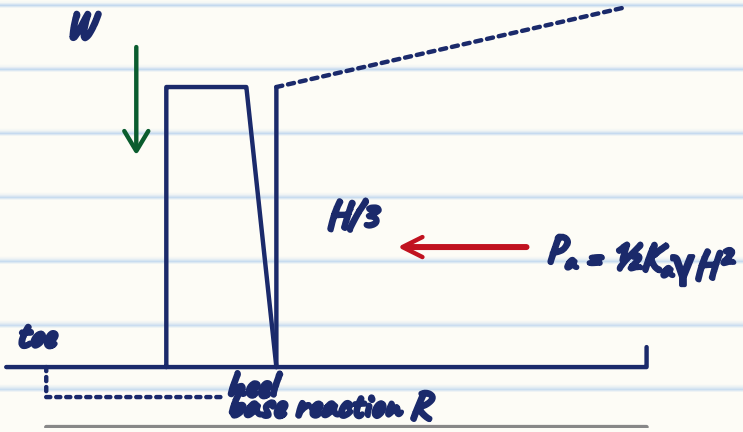


Fig 6 : Forces on a retaining wall

19. Stability Checks (per metre run)

(a) Overturning about toe

$$F.S._{overturning} = \Sigma M_R / \Sigma M_O = (W \cdot x + \ldots) / (P_h \cdot H/3) \geq 2.0$$

- ΣM_R = restoring moments of W, soil on heel (and P_v if any), about toe ; ΣM_O = overturning moment of horizontal thrust (and water). (IS 14458 : ≥ 1.5 with load factors; classical ≥ 2.)

(b) Sliding along base

$$F.S._{sliding} = (\mu \Sigma V + c_a B + P_p) / \Sigma H \geq 1.5$$

- $\mu = \tan \delta_b$ (≈ 0.5–0.6 for concrete on sand/gravel, $\tan \frac{2}{3}\varphi$ typically) ; c_a = adhesion (clay base). If F.S. < 1.5 → provide a **shear key** below base, increase base width, or tilt the base.

(c) Bearing pressure / tension at base

- Resultant ΣV acts at $\bar{x}$ from toe : $\bar{x} = (\Sigma M_R - \Sigma M_O) / \Sigma V$; eccentricity $e = B/2 - \bar{x}$

$$q_{max,min} = (\Sigma V / B)(1 \pm 6e/B) \text{ ; no tension if } e \leq B/6 \text{ (middle third)}$$

- $q_{max} \leq$ safe bearing capacity (F.S. ≥ 2.5–3 on q_{ult}) ; if $e > B/6$: $q_{max} = 2\Sigma V / (3\bar{x})$ (tension zone neglected).

(d) Other checks

- **Deep-seated (slope) stability** of the wall + soil mass (Unit II methods) ; F.S. ≥ 1.3–1.5.

- **Structural design** of stem, heel, toe slabs as cantilevers; for gravity walls check no tension in masonry at any section (for sections $e \leq b/6$).
- **Settlement** – tilt due to differential settlement (q_{max} at toe).

Proportioning of a cantilever wall (thumb rules)

- Base width $B \approx 0.5 - 0.7\ H$; base thickness $\approx H/10 - H/12$; stem thickness at base $\approx H/10 - H/12$, at top $\geq$ 200–300 mm; toe projection $\approx B/3$; embedment depth $\geq$ 1 m (below frost / scour).

Example 4 – Gravity Wall Stability

A masonry wall (γ = 22 kN/m^3) 4 m high, top width 1 m, base width 2.5 m, vertical back face, battered front. Backfill horizontal sand : γ = 18 kN/m^3, φ = 30°. Base friction μ = 0.5. Check stability (Rankine).

Item	Force (kN/m)	Arm from toe (m)	Moment (kN·m)
W_1 = 1 × 4 × 22 (rect.)	88	2.0	176
W_2 = ½ × 1.5 × 4 × 22 (tri.)	66	1.0	66
Σ	**154**		**242**

- K_a = 1/3 ; P_a = ½ × (1/3) × 18 × 4^2 = **48 kN/m** at $H/3$ = 1.33 m → M_0 = 48 × 1.333 = **64 kN·m**
- F.S. overturning = 242 / 64 = **3.78 > 2** ✔
- F.S. sliding = (0.5 × 154) / 48 = 77 / 48 = **1.60 > 1.5** ✔
- $\bar{x}$ = (242 – 64)/154 = 1.156 m ; e = 1.25 – 1.156 = **0.094 m < B/6 = 0.417** ✔ (no tension)
- q_{max} = (154/2.5)(1 + 6×0.094/2.5) = 61.6 × 1.226 = **75.5 kN/m²** ; q_{min} = 61.6 × 0.774 = **47.7 kN/m² ≤ SBC** ✔

Wall is safe against overturning, sliding and base tension; bearing OK if SBC $\geq \approx$ 75 kN/m² (checked against allowable).

Example 5 – Passive Resistance & Sliding with Key

A wall of ΣV = 200 kN/m and ΣH = 110 kN/m rests on sand with μ = 0.45. Find F.S. against sliding without and with a 0.8 m deep key having passive resistance $P_p = \frac{1}{2} K_p \gamma h^2$ (K_p = 3, γ = 18).

- Without key : F.S. = 0.45 × 200 / 110 = 90/110 = 0.82 < 1.5 ✘ unsafe
- P_p (key) = ½ × 3 × 18 × 0.8^2 = 17.3 kN/m (use ⅔ of it for safety ≈ 11.5)
- With key : F.S. = (90 + 17.3)/110 = 0.98 – still < 1.5 → **increase base width / add key / heel extension** ; each extra metre of heel adds soil weight (↑ ΣV).

Important Questions – Unit III

1. Explain active, passive and at-rest earth pressure. Give the order of K_a, K_0, K_p. (2/5 M)
2. State the assumptions of Rankine's theory. Derive K_a and K_p for cohesionless soil. (10 M)
3. Derive active pressure for c–φ soil; find depth of tension crack and critical height. (10 M)
4. Draw the pressure diagram for a layered backfill with water table and surcharge; find total thrust and point of application. (10 M)
5. Derive Coulomb's active earth pressure for a cohesionless backfill; show it reduces to Rankine. (10 M)
6. Explain Culmann's graphical method with a neat sketch. (5–10 M)
7. Explain Rebhann's graphical method for finding active earth pressure. (5–10 M)
8. Compare Rankine and Coulomb theories. (5 M)
9. Describe the different types of retaining walls with sketches. (10 M)
10. Explain the stability checks for a gravity / cantilever retaining wall. Numerical on overturning, sliding, base pressure. (10 M)

Formula sheet : $K_a = \tan^2(45^\circ - \varphi/2)$; $K_p = 1/K_a$; $z_0 = 2c/(\gamma\sqrt{K_a})$; $H_c = 4c/(\gamma\sqrt{K_a})$; $P_a = \frac{1}{2}K_a\gamma H^2$

All the best! Revise the figures – they carry marks.